AA 273 Midterm

Due no later than May 2 (Friday), 2025 at 3:30 PM Pacific on Gradescope

The midterm can be taken at any time during the 24 hour period it is available but must be completed within **a single 5 hour period**. The exam must be submitted no later than 3:30 PM (Pacific) on Friday, May 2nd through Gradescope.

Allowed resources

- Canvas (e.g., homework solutions, lecture notes/videos, starter code, past Ed posts)
- Textbooks
- Online documentation for any libraries that have been included in starter code (homework and exam). Examples include Numpy, Scipy, Matplotlib, Open3D, Pytorch, Pandas and standard Python libraries (math, csv).

Resources that are **NOT allowed**

- Material from previous iterations of the class (e.g., homework or exams)
- Discussion with classmates or anyone outside of the class
- The internet (except for documentation for the aforementioned libraries)
- Generative tools (including integrated coding tools like Copilot).
- You are not allowed to post questions relating to the midterm publicly on Ed between May 2 and May 5.

Additional Notes

- We prefer typeset solutions. However, if you choose to handwrite your solutions, the answers must be legible.
- Some questions require coding. Please provide associated code in your submission.
- If there are any questions regarding the exam, please direct them to the CAs through Ed as a private message or through email.

Q1 (20 points)

Bayes' Rule and Conditional Independence

You are working with a new medical sensor that can sense a subset of human emotion. Specifically, it can detect happiness, given by random variable X_1, and calmness, given by random variable X_2. $X_1 = 1$ indicates that a person is happy and $X_1 = 0$ indicates that the person is sad. Similarly, $X_2 = 1$ indicates that the person is calm, $X_2 = 0$ indicates that the person is agitated/excited. **Do NOT assume X_1 and X_2 are independent**. Table 1 gives the prior joint probability of X_1 and X_2.

This emotion sensor's measurement is given by Y and has two modes: green ($Y = 1$) or red ($Y = 0$). Table 2 gives the probabilities of the sensor's output given the person's emotional state. **Assume the measurement $Y^{(t)}$ at each time t is conditionally independent, conditioned on the states X_1 and X_2.**

Note: We do not consider natural dynamics of the system here (i.e., the person's emotion never changes), and there is no predict step.

X_1	X_2	$P(X_1, X_2)$
0	0	0.2
0	1	0.2
1	0	0.3
1	1	0.3

Table 1: Prior joint probability of X_1 and X_2

X_1	X_2	Y	$P(Y \mid X_1, X_2)$
0	0	0	0.9
0	0	1	0.1
0	1	0	0.8
0	1	1	0.2
1	0	0	0.4
1	0	1	0.6
1	1	0	0.05
1	1	1	0.95

Table 2: Sensor model $P(Y \mid X_1, X_2)$

(a) Express $P(Y = 1)$ as a function of known values, and compute the initial value using the prior values for X_1 and X_2.

(b) You receive one measurement $Y^{(1)} = \text{Green}$. What are your new probabilities $P(X_1 = 1 \mid Y^{(1)} = 1)$ and $P(X_2 = 1 \mid Y^{(1)} = 1)$?

(c) Generate the probability over time from $t_0 = 0$ to $t_f = 400$ of $P(X_1 = 1 \mid Y^{(1:t_f)})$ (being happy) and $P(X_2 = 1)$ (being calm), given the following measurement history:

- $Y^{(t)} = \text{Green}$ from $t = 1$ to $t = 100$
- $Y^{(t)} = \text{Red}$ from $t = 101$ to $t = 400$

Show $P(X_1 = 1 \mid Y^{(1:t_f)})$ and $P(X_2 = 1 \mid Y^{(1:t_f)})$ over time on the same plot, with clear axes labels, line labels, and a legend. You may use your coding language of preference. Submit your plot and your code.

Q2 [20 points]

Expectations, Means, Covariance, Moments, and related Statistical Concepts

Problem Setup: Consider an autonomous vehicle that is trying to estimate its position (denoted as X) using two different sensors. The true position, X, is unknown, and the vehicle receives two measurements: Z_1 and Z_2. Each sensor measurement consists of the true position plus some measurement noise:

$$Z_1 = X + V_1 \quad \text{and} \quad Z_2 = X + V_2$$

Where: $V_1 \sim \mathcal{N}(0, \sigma_1^2)$, $V_2 \sim \mathcal{N}(0, \sigma_2^2)$, and X, V_1, and V_2 are assumed to be mutually independent of each other.

You are tasked with developing an optimal linear estimator of X based on the two sensor measurements Z_1 and Z_2.

(a) Compute the means $\mathbb{E}[Z_1], \mathbb{E}[Z_2]$ and variances $\text{Var}(Z_1), \text{Var}(Z_2)$ of the two sensor measurements.

(b) Compute the covariance between the two measurements $\text{Cov}(Z_1, Z_2)$.

(c) Derive the expected value of the estimator $\hat{Z} = \alpha Z_1 + \beta Z_2$. What should be the relationship between α and β so that $\hat{Z}$ is an unbiased estimator of X. Hint: An estimator $\hat{\theta}$ of a parameter θ is called unbiased if $\mathbb{E}[\hat{\theta}] = \theta$, where $\mathbb{E}[\hat{\theta}]$ denotes the expected value of the estimator.

(d) What are the values of α and β so that $\hat{Z}$ has minimum variance and is an unbiased estimator of X? What is this minimum variance?

(e) Let this unbiased estimator of X with minimum variance be Z^*. Define the estimation error as $e = Z^* - X$

(i) What is the mean of this error $\mathbb{E}[e]$?

(ii) What is the variance of the error $\text{Var}(e)$?

(iii) Based on the noise characteristics of measurements, describe the probability distribution of the error e. What are the parameters of this distribution?

(iv) Using the distribution calculate the probability that the absolute error exceeds three standard deviations, i.e., compute: $P(|e| > 3\sigma_e)$

Express your answer in terms of the standard Gaussian cumulative distribution function: $\Phi(y) = P(Y \leq y)$, where $Y \sim \mathcal{N}(0, 1)$.

(f) Explain your intuition behind the obtained values of α and β for Z^*. Hint: What happens when one sensor is noisier than the other?

Q3 (20 points)

Dependence and Correlation

(a) Let $X = [X_1 \quad X_2]^T$ be a random variable described by a multivariate Gaussian distribution, $X \sim \mathcal{N}(\mu, \Sigma)$, where

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

Show that if $\Sigma_{12} = \mathbb{E}[(X_1 - \mu_1)(X_2 - \mu_2)^T] = 0$, X_1 and X_2 are independent.

(b) Let $W \in \mathbb{R}$ be a random variable that is uniformly distributed between interval $[0, 1]$, (i.e., $p(w) = 1$ and $\int_0^1 p(w)dw = 1$). Let V_1 and V_2 in $\mathbb{R}$ be two random variables that are normally distributed. $V_1 \sim \mathcal{N}(0, \sigma_1^2), V_2 \sim \mathcal{N}(0, \sigma_2^2)$. W, V_1, V_2 are all independent from each other. Consider the following two random variables:

$$Y_1 = \sin(2\pi W) + V_1$$

$$Y_2 = \cos(2\pi W) + V_2$$

Show mathematically that Y_1 and Y_2 are uncorrelated. Are they independent, why or why not? Are they conditionally independent given W? (Only a verbal justification is required for these last two questions.)

Q4 (20 points)

Modifications to the Kalman Filter

You are given a linear Gaussian model:

$$X_{t+1} = A_t X_t + B_t u_t + W_t$$
$$Y_{t+1} = C_t X_t + D_t u_t + V_t$$

Where W_t, V_t are Gaussian white noise processes, uncorrelated with each other and with the initial state X_0. In class, we derived the standard Kalman filter (KF) for the linear Gaussian model with some simplifying assumptions on the noise models W_t (process noise) and V_t (measurement noise), and set the matrix $D_t = 0$. In this problem, we will relax some assumptions to see how that changes our KF equations. **You can use the Gaussian estimation equations, as well as $Cov(X_t, V_t \mid Y_{1:t-1}) = 0$ and $Cov(X_t, W_t \mid Y_{1:t}) = 0$, or any other properties deriving from the noise independence and whiteness assumptions of the linear Gaussian model.**

(a) Derive the Kalman filter equations when the state-space matrix $D_t \neq 0$. Which equations are the same and which are different from the original KF?

(b) We previously assumed that the process noise and the measurement noise have zero mean $\mu_{W_t} = \mu_{V_t} = 0 \; \forall t$. You can assume $D_t = 0$ for this part. Now, derive the Kalman filter equations when

$$W_t \sim \text{GWN}\left(\mu_{W_t}, Q\right), \quad \mu_{W_t} \neq 0$$
$$V_t \sim \text{GWN}\left(\mu_{V_t}, R\right), \quad \mu_{V_t} \neq 0$$

Which equations are the same and which are different from the original KF?

Let us now apply the result from (b) to a simulation environment. **You are given the same simulation environment as the drone in HW3.** Let $p_t \in \mathbb{R}^2$ be its position, and $s_t \in \mathbb{R}^2$ its velocity. Let's model its dynamics as a discrete time double integrator perturbed by noise,

$$P_{t+1} = P_t + \delta t S_t \tag{1}$$
$$S_{t+1} = S_t + \delta t u_t + W_t, \tag{2}$$

where $u_t \in \mathbb{R}^2$ is the acceleration commanded by a remote human pilot, and δt is the discrete time step. The drone makes position measurements with GPS according to the noisy measurement model

$$Y_t = P_t + V_t. \tag{3}$$

The processes W_t, V_t are Gaussian white noise with means μ_W and μ_V, covariances Q and R, respectively, uncorrelated with each other and uncorrelated with the initial state. Let $\delta t = 1s$, $Q = I_2 m^2/s^2$, $R = 9\text{I}_2 m^2$, $\mu_W = [0.3, 0.3]^\top m/s$, $\mu_V = [400, 400]^\top m$, $u_t = -2.5[\cos(0.05t) \;\; \sin(0.05t)]^T m/s^2$.
Hint: You can use the `numpy.random.multivariate_normal()` *function to create the noise.*
Note: For consistency in solutions, seed your random variables using `np.random.seed(273)`

(c) Suppose your initial guess about the state of the drone has a mean $\mu_0 = [1500m \quad 100m \quad 0m/s \quad 55m/s]^T$, and covariance matrix $\Sigma_0 = [250000\text{I}m^2 \quad 0; \quad 0 \quad \text{I}m^2/s^2]$. Use the Kalman filter equations you derived in (b) to find your estimated state trajectory. As in PS3, use your error ellipse script from PS2 to plot the means and error ellipses of the position estimate over the top of the true trajectory.